

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
# Importing Data
data = pd.read_csv('/content/housing.csv')
```

```
data.head()
data.shape
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 505 entries, 0 to 504
Data columns (total 1 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0    0.00632  18.00    2.310  0  0.5380  6.5750  65.20  4.0900  1  296.0  15.30  396.90  4.98  24.00  505 non-null  object
dtypes: object(1)
memory usage: 4.1+ KB
```

```
data = pd.read_csv(
    '/content/housing.csv',
    delim_whitespace=True,
    header=None
)
```

```
/tmp/ipython-input-1202571515.py:1: FutureWarning: The 'delim_whitespace' keyword in pd.read_csv is deprecated and will be removed
data = pd.read_csv(
```

```
data.columns = [
    'CRIM', 'ZN', 'INDUS', 'CHAS', 'NOX', 'RM',
    'AGE', 'DIS', 'RAD', 'TAX', 'PTRATIO',
    'B', 'LSTAT', 'PRICE'
]
```

```
data.head()
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 506 entries, 0 to 505
Data columns (total 14 columns):
#   Column      Non-Null Count  Dtype
---  -
0    CRIM        506 non-null  float64
1    ZN          506 non-null  float64
2    INDUS       506 non-null  float64
3    CHAS        506 non-null  int64
4    NOX         506 non-null  float64
5    RM          506 non-null  float64
6    AGE         506 non-null  float64
7    DIS         506 non-null  float64
8    RAD         506 non-null  int64
9    TAX         506 non-null  float64
10   PTRATIO     506 non-null  float64
11   B           506 non-null  float64
12   LSTAT       506 non-null  float64
13   PRICE       506 non-null  float64
dtypes: float64(12), int64(2)
memory usage: 55.5 KB
```

```
# Input features
x = data.drop('PRICE', axis=1).values

# Output target
y = data['PRICE'].values
```

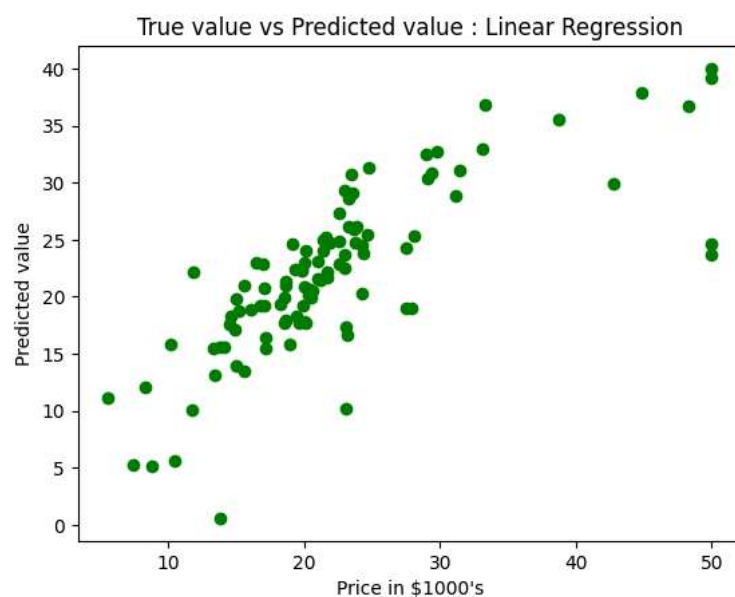
```
from sklearn.model_selection import train_test_split
xtrain, xtest, ytrain, ytest = train_test_split(
    x, y, test_size=0.2, random_state=0
)

from sklearn.linear_model import LinearRegression
regressor = LinearRegression()
regressor.fit(xtrain, ytrain)

y_pred = regressor.predict(xtest)
```

```
plt.scatter(ytest, y_pred, c='green')
plt.xlabel("Price in $1000's")
plt.ylabel("Predicted value")
plt.title("True value vs Predicted value : Linear Regression")
plt.show()

from sklearn.metrics import mean_squared_error, mean_absolute_error
print("MSE:", mean_squared_error(ytest, y_pred))
print("MAE:", mean_absolute_error(ytest, y_pred))
```



MSE: 33.44897999767653
MAE: 3.8429092204444966

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